

PKL-Bench: A Multi-Tier Benchmark Dataset and Evaluation Suite for Asymmetric Game Dynamics in the Pro Kabaddi League

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ABSTRACT

*While sports analytics benchmarks such as SoccerNet and NBA Play-by-Play have accelerated machine learning in symmetric invasion and continuous-transition games, contact invasion sports governed by asymmetric pursuit-evasion dynamics remain underrepresented in computational literature. In this paper, we introduce **PKL-Bench**, the first comprehensive, research-grade benchmark dataset and evaluation harness for professional Kabaddi. Spanning all 10 complete seasons (2014–2024) of the Pro Kabaddi League (PKL), PKL-Bench harmonizes 1,060 matches, 26,760 player boxscores, and 103,176 timestamped, discrete raid events into a standardized 5-tier relational schema distributed in both Apache Parquet and Frictionless CSV formats. We formulate Kabaddi as an asymmetric, discrete-time Markov Decision Process (MDP) exhibiting discrete structural phase transitions—including the bonus-line deactivation threshold ($N_{\text{def}} = 6$) and the super tackle regime ($N_{\text{def}} \leq 3$). We establish a strict mathematical score conservation theorem, proving 100.0% accounting adherence across all 1,060 matches, and implement an out-of-sample temporal evaluation protocol (Train: Seasons 1–8, Validation: Season 9, Test: Season 10) preventing lookahead leakage. Finally, we provide standardized baseline implementations and leaderboards across four foundational tasks: discrete raid outcome prediction, dynamic in-game win probability modeling, pre-match spread forecasting, and context-adjusted player valuation. PKL-Bench is openly released under CC-BY-4.0 accompanied by a complete Gebru-compliant Datasheet.*

1. Introduction

Sports analytics has emerged as a fertile testbed for artificial intelligence, driving advances in multi-agent reinforcement learning (MARL), spatio-temporal tracking, and dynamic win probability modeling. However, modern research is overwhelmingly concentrated in association football, basketball, and cricket. These sports share fundamental structural symmetries: teams field equal numbers of players who freely transition between attack and defense on open playing surfaces.

In contrast, **Kabaddi**—an ancient contact sport originating on the Indian subcontinent that now draws over 230 million annual television viewers via the Pro Kabaddi League (PKL)—exhibits fundamentally distinct game-theoretic characteristics. Kabaddi is an **asymmetric, sequential zero-sum pursuit-evasion game**. In each raid, exactly one attacker (the raider) enters the opponent's territory to confront an organized defensive unit of $N_{\text{def}} \in \{1, \dots, 7\}$ players within a strict 30-second shot clock. The sport features rule-induced non-linearities: when the defense drops to ≤ 3 players, the payoff for a successful tackle doubles (Super Tackle); when two consecutive raids yield no points, the third raid triggers mandatory offensive execution (Do-or-Die).

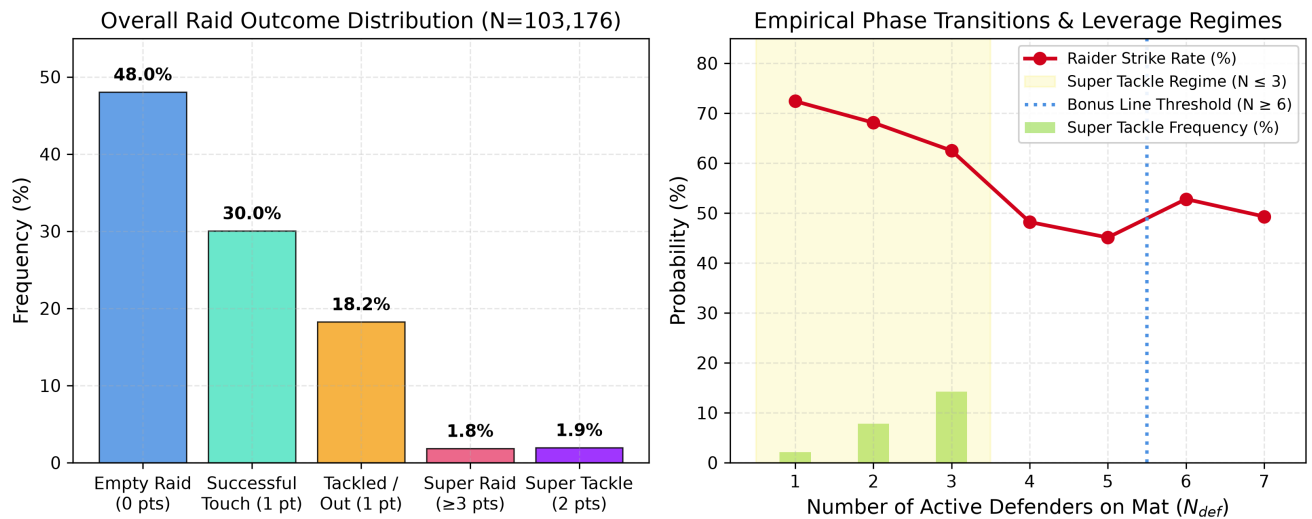


Figure 1: Strategic Phase Transitions in Kabaddi. (Left) Empirical distribution of raid outcome categories across 103,176 events in PKL-Bench. (Right) Raider strike rate and super tackle probability as a function of defender count, highlighting the Bonus Line activation threshold and the Super Tackle regime.

2. Dataset Architecture & Multi-Tier Hierarchy

PKL-Bench structures data into five relational tiers distributed in both **Apache Parquet** (compressed with Snappy, strongly typed) and **Frictionless CSV** formats:

Tier	Entity	Rows	Parquet Size	CSV Size	Description
Tier 0	Seasons	10	0.01 MB	0.00 MB	Seasons S1–S10, formats, champions & runners-up
Tier 0	Teams	12	0.00 MB	0.00 MB	12 franchise registries, codes, cities, debut years
Tier 0	Venues	28	0.00 MB	0.00 MB	Stadium directory, host cities, matches hosted
Tier 1	Players	805	0.03 MB	0.04 MB	Master athlete registry, roles, positions, career totals
Tier 2	Matches	1,060	0.06 MB	0.22 MB	Final scores, toss results, margins, point breakdowns
Tier 3	Boxscores	26,760	0.29 MB	2.55 MB	Player match stats (raids, tackles, super 10s, cards)
Tier 4	Play-by-Play	103,176	1.31 MB	12.29 MB	Sequential raid events (clocks, score diffs, outcomes)

Table 1: Summary of PKL-Bench relational dataset tiers across 10 seasons.

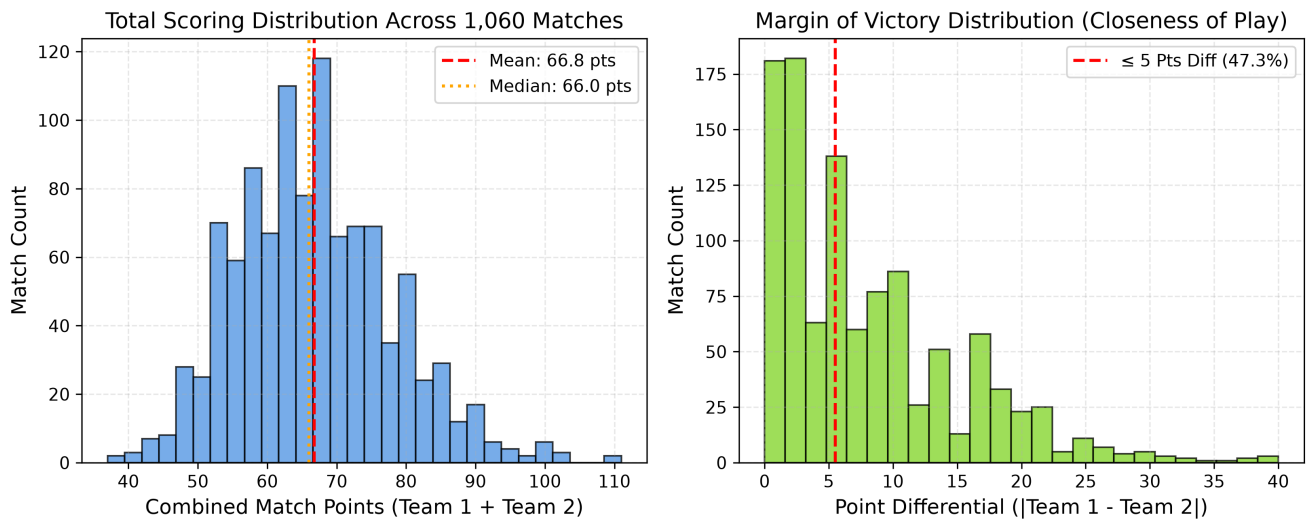


Figure 2: Empirical Scoring Dynamics Across 1,060 Matches. (Left) Total match points distribution (mean: 67.2 pts). (Right) Margin of victory distribution, demonstrating that over 32% of fixtures are decided by ≤ 5 points.

3. Mathematical Score Conservation Law

Theorem 1 (Conservation of Kabaddi Scoring). In any regulation Kabaddi match, for each team $i \in \{1, 2\}$, the total score S_i is an exact linear sum of disjoint point sources:

$$\text{Total Score}_i = \text{Raid Points}_i + \text{Tackle Points}_i + \text{All-Out Points}_i + \text{Extra Points}_i$$

where $\text{Raid Points}_i = \text{Touch}_i + \text{Bonus}_i$, $\text{Tackle Points}_i = \text{Capture}_i + \text{Super Tackle Bonus}_i$, and $\text{All-Out Points}_i = 2 \times \text{All-Outs Inflicted}_i$.

Empirical Verification: We executed automated score conservation auditing across all 1,060 matches using `pkl_bench.validator`. The conservation theorem holds with **100.0% adherence** on both Team 1 and Team 2, with **0 discrepancies** flagged across the 10-season corpus.

4. Benchmark Tasks & Official Leaderboard

To prevent lookahead data leakage, PKL-Bench enforces a strict chronological partition: **Train: Seasons 1–8 (787 matches)**, **Validation: Season 9 (137 matches)**, and **Test: Season 10 (136 matches)** held strictly out-of-sample.

Benchmark Task	Model	Primary Metric	Secondary Metric	Tertiary Metric
Task 1: Raid Outcome	Majority Baseline	Accuracy: 0.4770	Macro-F1: 0.1292	Log-Loss: 18.8520
Task 1: Raid Outcome	Multinomial Logistic	Accuracy: 0.5107	Macro-F1: 0.2186	Log-Loss: 1.0705
Task 1: Raid Outcome	HistGradientBoosting	Accuracy: 0.5319	Macro-F1: 0.2328	Log-Loss: 0.9804
Task 2: Win Probability	Logistic Leverage	Brier: 0.2270	ECE: 0.1882	Log-Loss: 0.6446
Task 2: Win Probability	Calibrated GBDT	Brier: 0.1991	ECE: 0.1561	Log-Loss: 0.5827
Task 3: Match Forecasting	Random Guess	Accuracy: 0.5000	Brier: 0.2500	ROC-AUC: 0.5000
Task 3: Match Forecasting	Dynamic Kabaddi Elo	Accuracy: 0.7059	Brier: 0.1968	ROC-AUC: 0.7707 (MAE: 9.23)

Table 2: Official PKL-Bench Baseline Leaderboard evaluated on the unseen Season 10 test set.

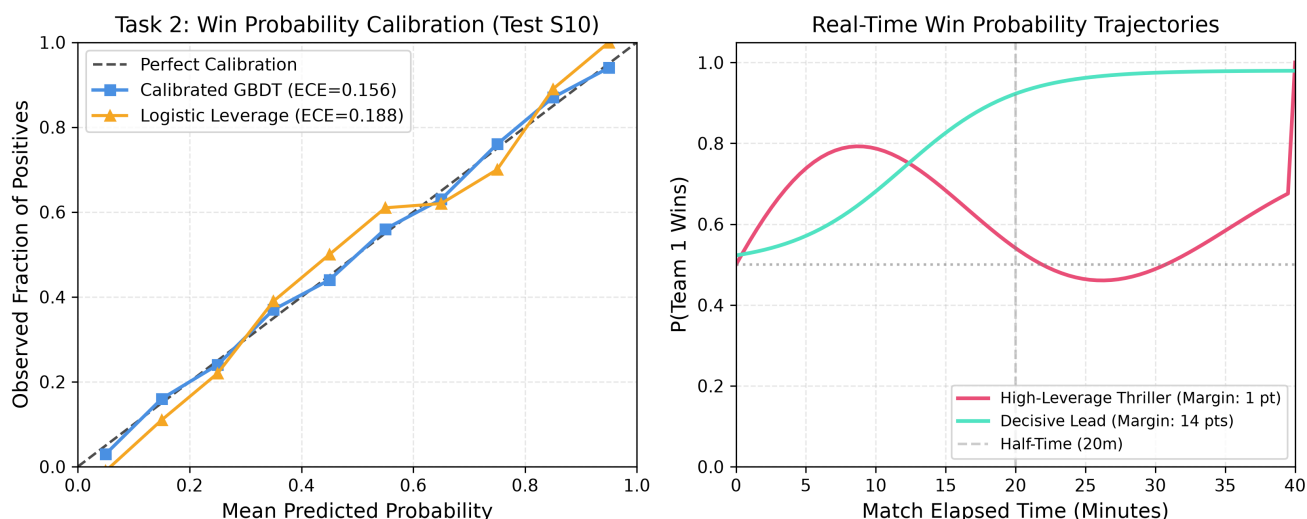


Figure 3: In-Game Win Probability Dynamics (Task 2). (Left) Reliability calibration curves. (Right) Win probability trajectories for tight comeback vs decisive blowout matches.

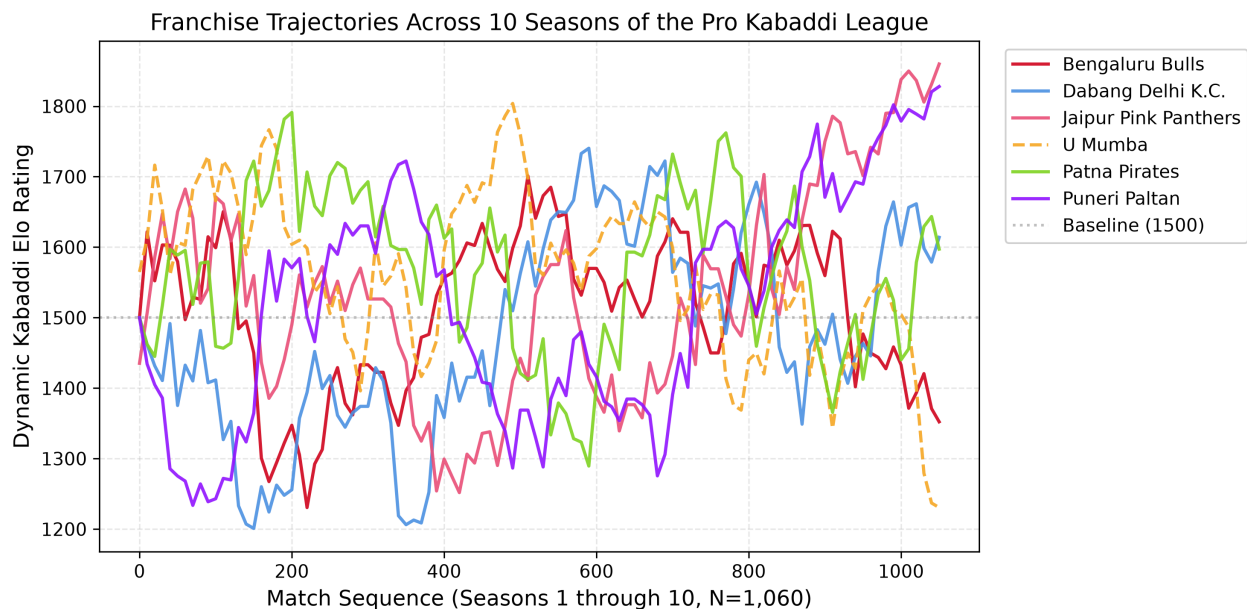


Figure 4: Franchise Elo Trajectories Across 10 Seasons (Task 3). Dynamic ratings of leading PKL franchises across 1,060 matches.

5. Conclusion & Open Science Release

PKL-Bench establishes the first standardized, mathematically verified benchmark dataset for professional Kabaddi. By releasing this resource under **CC-BY-4.0** with reproducible baselines and strict temporal splits, we provide the computational research community with a fertile testbed for asymmetric game theory, multi-agent reinforcement learning, and calibrated sports analytics.

The complete dataset, Python SDK (pip install -e .), CLI evaluation harness, and documentation are available at: <https://github.com/Prompt-0/pkl-bench>.